

Pre-Export Row Scaling for Generated Mixed-Integer Programs: Mechanism Attribution and Residual-Budget Contracts

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Abstract Multiplying a row of a mixed-integer program by a positive number preserves it exactly and changes the numbers a solver sees. We ask what that change buys and what it still guarantees. On 306 generated hydrothermal-dispatch models, holding each model's LP basis fixed across its representations, geometric-mean row scaling lowers the condition number of the exported file on every model, by six to eight orders of magnitude in the median of each class. That figure describes the file: rescaling the same bases the way a solver does recovers most of it, leaving a factor of a few, and on one class it inverts the ordering of kernels, so a diagnostic read on the exported matrix does not rank scaling policies. We then separate two kinds of information. An apparent block-metadata effect is a coupling-kernel effect, because geometric-mean centering makes the extrema-based block statistic identically one; a residual budget in application units, by contrast, survives consistent re-emission, and we characterize exactly the row factors a budget and coefficient limits admit. In reconstructed dispatch runs, scaling used 30–32% less solver work on the instances every policy completed, while one timed out under every scaled policy and reversed the failure-sensitive ordering. In an enumerated stress test, plain centering lost verified optima on all three solvers and adding the budget to that same rule lost none. A representation can be numerically better without being operationally safer, unless what safer means is stated in the units of the original model.

Keywords Mixed-integer linear programming · Row scaling · Numerical reliability · Equilibration · Residual budgets · Computational reproducibility

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1 Introduction

A model generator can emit the same mixed-integer linear program (MILP) in many mathematically equivalent forms. Multiplying a constraint and its right-hand side (RHS) by a positive number preserves the MILP in exact arithmetic, yet it changes the coefficients on which presolve and LP factorizations operate and the original-unit meaning of an absolute residual tolerance. Solvers scale internally; a generator can also scale before export and use information that a matrix file does not carry, such as row roles, variable ownership and application units [15, 17].

The numerical effect of external row scaling looks large. On 306 dispatch models from a hydrothermal generator, holding one LP basis fixed across representations, geometric-mean row scaling lowers the 1-norm condition number of that basis on every model, by six to eight orders of magnitude in the median of each class (Section 7.1). It looks large because it is measured on the file the generator writes. Rescale the same bases the way a solver does and most of it is recovered by the solver itself, leaving a factor of a few, and on one of the three classes the ordering of the scaling kernels reverses (Section 7.2). So the number is real, it is not the whole story, and a gain of this kind leaves one question open, in two halves:

when external preprocessing changes a generated MILP, what caused the numerical improvement, and what guarantee about the original model survives the change?

Both halves are about the same thing, the information the generator holds: which part of it explains the gain, and which part of it can be turned into a guarantee. The first half is answered by identifying the mechanism, the second by stating a contract in the units of the original model and checking it against what a solver returns. Representation quality, solver effort, failure-sensitive performance and verified accuracy are different endpoints, and evidence for one does not transfer to another.

This paper answers both halves for external row scaling of generated MILPs. Its contributions are the following.

- (A) *Attribution.* A comparison identifies the effect of block ownership only when coverage, row roles, kernels and safeguards are matched. With such controls, the order-of-magnitude conditioning advantage of a block-relative coupling rule over flat geometric-mean normalization is reproduced by a rule that uses no block scales, including its growth with block heterogeneity (Sections 4 and 7.3). A matrix diagnostic read on the exported file also fails to rank the kernels it compares: recomputed after the scaling a solver applies, the separation falls from orders of magnitude to a factor of a few and, on one class, changes sign (Section 7.2).
- (B) *Mechanism.* Geometric-mean row centering makes the post-local extrema-based block statistic identically one, so the block-relative coupling formula reduces to Euclidean row normalization. After local whitening — that is, once each local block’s rows have been normalized so that the block is an

identity as far as scale is concerned — the idealized coupling problem has the closed-form optimal common factor $(1 + \rho^2)^{-1/2}$, which requires no ownership metadata (Section 4).

- (C) *Contract*. Residual budgets stated in application units survive consistent row re-emission, that is, rewriting a row and its right-hand side in different units. We characterize the admissible row factors exactly, with necessary and sufficient compatibility conditions, an incompatibility certificate, the constrained logarithmic minimax factor, the integer-rounding term that scaling cannot remove, and exact binary64 export with power-of-two factors (Section 5).
- (D) *Validation*. A fixed-basis audit of 306 dispatch models, recomputed through a solver-style equilibration, a fresh audit of 1,080 matched matrix exports, a reconstruction of 240 operational solver runs with two matched sensitivity reruns of the same size, and an exactly enumerated stress experiment with 4,752 optimizer calls on three solvers (Sections 6 and 7).

Contributions (A) and (B) answer the first half of the question, (C) the second, and (D) is the evidence for both. The results progress from the representation to the solver. Scaling changes the numerical representation by orders of magnitude, and the kernel determines how much. The apparent metadata effect is a kernel effect with a simpler explanation. Residual budgets have measurable consequences: adding one to a centering rule never lost a verified stress configuration, relative to that same unbudgeted rule, on any of the three solvers tested. Whether a solver accepts the resulting points, and how much effort it spends, remain properties of the solver pipeline, to be measured and verified in original units.

2 Related work

Diagonal scaling has a long history. Bauer [2] posed optimal scaling as a minimization of the condition number, van der Sluis [27] proved that equilibrating rows in the infinity norm minimizes κ_∞ over all row scalings and that normalizing rows in the Euclidean norm comes within $\sqrt{n}$ of the best κ_2 , and Tomlin [28] examined what these results mean for linear programs; Higham [12, Ch. 9] surveys the perturbation theory. Curtis and Reid [6] give the logarithmic least-squares method, Elble and Sahinidis [9] compare scalers for simplex and find that additional complexity need not improve performance, and Klotz [20] relates numerical instability to formulation and finite precision. Optimal diagonal preconditioning is a broader problem: Qu et al. [26] use quasiconvex and semidefinite formulations, while Demmel [7] gives block-Jacobi guarantees.

The classical results already explain why a Euclidean kernel competes with a geometric-mean one, and that is the qualitative content of our attribution result. What they do not give is the object we need: the optimal value of the single factor that couples two normalized blocks, and how that optimum interacts with an admissibility interval imposed from outside. Our triangular calculation supplies both in closed form, and the attribution experiment is

what shows that the quantity a structure-aware policy believed it was using had already been removed by its own preprocessing.

Robustness is also an established objective in MIP scaling. Berthold and Hendel [3] learn to select between standard and Curtis–Reid scaling in Xpress from an indicator of numerical difficulties. Their implementation uses power-of-two factors and their evaluation examines inconsistencies across seeds. Our external construction instead states row-level admissibility conditions; it does not predict or replace internal solver scaling.

Modeling environments already support generator-level information: Pyomo’s `ScaleModel` applies component factors [25], GAMS has variable and equation scale attributes [10], and IDAES derives and propagates scales across submodels [16, 17]. Plasmo.jl retains local problems and linking constraints [19]. Consequently, using modeling metadata before export is not new. Our identifiability result concerns the loss of one specific block statistic after a preceding normalization.

Tolerance-based scaling also has direct precedent. Gurobi documents absolute feasibility and integrality tolerances and the treatment of very small coefficients [11], and recommends external scaling that makes tolerance-sized violations negligible in application units [15]. Pritchett, Howell and Grebow [24] similarly scale constraints using physical accuracy requirements and SNOPT’s tolerance. Section 5 combines this known transfer with coefficient limits, identifies incompatible specifications, and solves the constrained centering problem. Integer rounding and exact binary64 export impose additional restrictions.

Integer-aware presolve can instead strengthen the formulation. GCD division and RHS rounding are established reductions [1]; on integer rows they can remove a narrow feasibility margin that pure scaling preserves. We include that distinction in the stress comparison, rather than treating tolerance-based scaling as a replacement for presolve.

Finally, correctness under finite precision has its own line. Neumaier and Shcherbina [23] build small mixed-integer programs on which tolerance-based solvers return wrong answers, and give safe bounds that avoid it; Miltenberger, Ralphs and Steffy [22] examine where numerical trouble arises inside branch-and-cut. Exact rational solvers remove the question entirely [5, 8], at a cost that our enumerated family does not have to pay but that larger instances would. Cheung, Gleixner and Steffy [4] provide certificates for integer-programming results, and Hoen and Gleixner [14] analyze individual branch-and-bound decisions. Our rational residual checks address primal acceptance, and independent enumeration supplies optimality on the small binary instances. What this paper adds to these lines of work is attribution: matched comparisons of actual exports that separate what a scaling policy changes from which information causes the change.

Table 1 Policies. Coverage and safeguards are matched across the comparisons in which a policy is used; a role map and a block-scale map are separate information sources. The block scales s_i are defined in Section 4.

Policy	Local rows	Coupling rows	Information used
Base	None	None	None
Flat-GM	GM	GM	Row coefficients
Flat-L2	L2	L2	Row coefficients
Role-Hybrid	GM	L2	Row roles, coefficients
SA-Mat	GM	$\left[\sum_j (a_{rj}/s_{\beta(j)})^2\right]^{-1/2}$	Roles, block scales, ownership
SA-Pre	GM	Same expression, pre-local scales	Roles, original block scales
SA-Mat-permB	GM	Same expression, permuted block scales	Roles, permuted ownership

3 Model, policies and comparisons

Consider

$$\min c^T x \quad \text{subject to } Ax \leq b, \quad l \leq x \leq u, \quad x_j \in \mathbb{Z} \ (j \in J),$$

where equalities and opposite inequality senses are allowed. For a positive diagonal matrix D_r , replacing (A, b) by $(D_r A, D_r b)$ preserves every constraint, variable domain and objective value in exact arithmetic. The whole row, including its RHS, is multiplied by one positive number; dividing different coefficients of a row by different block scales is not a row scaling and is outside this equivalence.

The generator provides a partition of the local rows into nonempty blocks, a set of coupling rows, and a column-to-block map β . A scaling policy is then determined by four separate choices: *coverage*, the rows eligible for scaling; the *kernel*, which maps an eligible row to a candidate factor; *safeguards*, which decide whether a candidate is applied; and *information*, the generator data the kernel may read, such as roles, block scales and ownership. An effect can be attributed to information only when the other three choices are matched.

For a nonzero row, put $m_r = \min_{a_{rj} \neq 0} |a_{rj}|$ and $M_r = \max_j |a_{rj}|$. The geometric-mean (GM) factor is $(m_r M_r)^{-1/2}$ and the Euclidean (L2) factor is $1/\|a_r\|_2$. Both are coefficient-only, but they are different kernels, and Table 1 keeps the distinction visible.

The last policy is a control rather than a proposal: SA-Mat-permB applies the SA-Mat formula with the block scales assigned to the wrong blocks. Relabeling blocks without changing their assignment is an invariance check, whereas permuting the assignment of unequal block scales tests whether the map is useful on the chosen instances.

4 Identifying the mechanism

This section shows that, under exact geometric-mean local normalization, the extrema-based post-local block statistic used by the original policy carries no block-scale information, so the coupling stage cannot use it, and it derives the simpler mechanism that remains.

4.1 Collapse of post-normalization block centers

For each nonzero local row r , write m_r and M_r for its smallest and largest nonzero coefficient magnitudes. The geometric-mean kernel applies $d_r = (m_r M_r)^{-1/2}$. Let $\tilde{m}_r = d_r m_r$ and $\tilde{M}_r = d_r M_r$ denote the exported extrema. For a nonempty set I of local rows, the block statistic is

$$s_I = \sqrt{\left(\min_{r \in I} \tilde{m}_r\right) \left(\max_{r \in I} \tilde{M}_r\right)}.$$

Proposition 4.1 (Exact collapse and the effect of a skip band) *If every row receives its geometric-mean factor, then $s_I = 1$ for every nonempty grouping I of local rows. More generally, suppose the only safeguard replaces d_r by 1 when the candidate lies in $(1/u, u)$, where $u \geq 1$. Then $s_I \in [1/u, u]$, independently of the original coefficient scales and of the grouping of the rows.*

Proof Under exact normalization, $\tilde{m}_r = \sqrt{m_r/M_r} = 1/\tilde{M}_r$. Consequently $\min_r \tilde{m}_r = 1/\max_r \tilde{M}_r$, proving the first statement. For the second, let $c_r = \sqrt{\tilde{m}_r \tilde{M}_r}$. An accepted row has $c_r = 1$; a skipped row has $c_r = \sqrt{m_r M_r} \in (1/u, u)$. Thus $u^{-2} \leq \tilde{m}_r \tilde{M}_r \leq u^2$ for every row. Choose p attaining the maximum exported row magnitude and q attaining the minimum. Then

$$u^{-2} \leq \tilde{m}_q \tilde{M}_q \leq \tilde{m}_q \tilde{M}_p \leq \tilde{m}_p \tilde{M}_p \leq u^2.$$

Taking square roots gives the result.

For a coupling row a_r , let $\beta(j)$ assign column j to a local block, and define the block-weighted candidate and its ownership-blind counterpart by

$$\gamma_r^{\text{block}} = \left(\sum_j a_{rj}^2 / s_{\beta(j)}^2 \right)^{-1/2}, \quad \gamma_r^{\text{L2}} = \left(\sum_j a_{rj}^2 \right)^{-1/2}.$$

Blockless columns may use an arithmetic mean of valid block scales or the unit fallback; both conventions preserve the interval in Proposition 4.1.

Proposition 4.2 (Loss of ownership information) *Under exact local normalization and with no coefficient filtering, $\gamma_r^{\text{block}} = \gamma_r^{\text{L2}}$ for every coupling row, so changing the ownership map cannot change the candidate export;*

applying the same coupling safeguards to identical candidates preserves equality. Under the skip-band assumptions of Proposition 4.1,

$$u^{-1} \leq \frac{\gamma_r^{\text{block}}}{\gamma_r^{\text{L2}}} \leq u,$$

and, before coupling safeguards,

$$u^{-2} \leq \frac{\kappa_2^+(A_{\text{block}})}{\kappa_2^+(A_{\text{L2}})} \leq u^2,$$

where $\kappa_2^+(A) = \sigma_{\max}(A)/\sigma_{\min}^+(A)$ is taken over the positive singular values, so the bound covers rectangular and rank-deficient exports.

Proof The equality follows by substituting $s_i = 1$. In the second case,

$$u^{-2} \|a_r\|_2^2 \leq \sum_j a_{rj}^2 / s_{\beta(j)}^2 \leq u^2 \|a_r\|_2^2,$$

which gives the factor bound. Write the block-based export as EA_{L2} , where E has unit entries on local rows and entries in $[1/u, u]$ on coupling rows. Multiplication by E changes each singular value by a factor between $1/u$ and u and preserves rank, so $\sigma_{\max}$ and $\sigma_{\min}^+$ each move by at most u ; taking the two extremes in opposite directions gives the stated interval for their ratio.

Both statements assume that the block summary sees every coefficient. Coefficient-window vetoes and filtered extrema lie outside the hypotheses: a row with magnitudes $(10^{-8}, 3, 10^8)$ is geometrically centered, but restricting the summary to $(10^{-6}, 10^6)$ produces $s_I = 3$, a nonunit statistic created by the filter rather than by retained physical scale. The export audit in Section 7 measures how often the implemented safeguards break exact equality.

4.2 What a matched comparison identifies

With equal coverage and equal local kernels, two exports agree on their local rows, so any difference is confined to the coupling rows. This localizes the difference without attributing it to ownership, because the coupling kernels may still differ. The ownership control therefore keeps local GM and coupling L2 normalization but replaces every block scale by one (Role-Hybrid in Table 1); SA-Mat versus Role-Hybrid is the ownership comparison. Flat-GM changes the coupling kernel and Flat-L2 changes the local kernel, so neither isolates ownership.

Equality also requires equal *applied* factors, not merely an inactive coupling stage in one policy. For local rows $x_1 \leq 1$, $x_2 \leq 1$ and the coupling row $x_1 + 10^{-2}x_2 \leq 1$, both block scales are one and the Euclidean coupling candidate $(1 + 10^{-4})^{-1/2}$ falls inside the skip band $(1/2, 2)$, whereas the flat GM candidate 10 is applied. Both policies cover every row, yet their exports differ.

4.3 The spectral slope in a stylized model

Take $B \geq 2$ local singleton rows with positive coefficients $v_i = 10^{e_i}$, where $\min_i e_i = -R$ and $\max_i e_i = R$, and one coupling row $v^\top$. Exact local normalization produces I_B and unit post-local block centers. A whole-row coupling factor γ gives

$$A_\gamma = \begin{bmatrix} I_B \\ \gamma v^\top \end{bmatrix}, \quad A_\gamma^\top A_\gamma = I_B + \gamma^2 v v^\top, \quad \kappa_2(A_\gamma) = \sqrt{1 + \gamma^2 \|v\|_2^2}.$$

Flat geometric-mean normalization selects $\gamma = 1$, whereas both the post-local block rule and the hybrid Euclidean control select $\gamma = 1/\|v\|_2$. Hence

$$\kappa_2(A_{\text{GM}}) = \sqrt{1 + \sum_i 10^{2e_i}}, \quad \kappa_2(A_{\text{block}}) = \kappa_2(A_{\text{hybrid}}) = \sqrt{2}.$$

For fixed B the ratio is of order 10^{-R} : the advantage grows with block heterogeneity and survives removal of the ownership map. Every coupling coefficient is multiplied by the same γ , so all within-row ratios are preserved; dividing each v_i by its own 10^{e_i} would change the constraint and is not a row scaling.

4.4 Optimal coupling factor after local whitening

The stylized model suggests a constructive replacement for the metadata explanation: once local scale has been removed, choose the coupling factor from the coupling coefficients themselves. In the idealized triangular model this choice has an exact solution.

Proposition 4.3 (Optimal common coupling factor after local whitening) *Let $R \in \mathbb{R}^{m \times n}$, with $m, n \geq 1$, and set $\rho = \|R\|_2$. For*

$$C_\gamma = \begin{bmatrix} I_n & 0 \\ \gamma R & \gamma I_m \end{bmatrix}, \quad \gamma > 0,$$

the unique factor minimizing $\kappa_2(C_\gamma)$ is

$$\gamma_* = (1 + \rho^2)^{-1/2}, \quad \min_{\gamma > 0} \kappa_2(C_\gamma) = \sqrt{1 + \rho^2} + \rho.$$

On an admissible interval $0 < \ell \leq \gamma \leq h$, the optimum is $\min\{h, \max\{\ell, \gamma_\}\}$.*

The proof (Online Resource 1) reduces the spectrum by an SVD to two-by-two blocks. For the block at ρ , the singular-value ratio k satisfies $k + k^{-1} = \gamma^{-1} + \gamma(1 + \rho^2)$, whose unique minimizer gives the result.

For a single coupling row, γ_* is the Euclidean normalization of the complete row, including the coefficient of its aggregate variable. The idealized coupling problem therefore has a simple optimal whole-row normalization that uses no ownership information, and this is the mechanism the matched control isolates

empirically. The statement concerns a common factor on a locally whitened triangular structure; for general generated MIPs it motivates the Euclidean coupling kernel rather than certifying it.

That a Euclidean rule should do well is not itself news: van der Sluis [27] showed that normalizing rows in the Euclidean norm comes within $\sqrt{n}$ of the best κ_2 attainable by any row scaling, which is already enough to expect the control to be competitive. Two things here are not supplied by that classical bound, and both are what the attribution argument needs. The first is an exact optimum rather than a factor: on this structure the best common factor is $(1 + \rho^2)^{-1/2}$ and the best attainable condition number is $\sqrt{1 + \rho^2} + \rho$, so there is nothing left for a block-scale map to contribute, whereas a $\sqrt{n}$ bound would leave room for one. The second is the behaviour of that optimum under an external restriction: the interval form below is what connects this section to the contract of Section 5, and a bound on the attainable condition number says nothing about it.

Why conditioning needs a residual bound. The interval form of Proposition 4.3 matters. In A_γ , κ_2 decreases to one as $\gamma \rightarrow 0$, while a fixed residual on the exported coupling row corresponds to an original residual larger by the factor $1/\gamma$. A conditioning criterion alone cannot say how small a coupling factor may be: on any interval its minimizer for A_γ is the lower endpoint ℓ . Section 5 derives that endpoint from an application-unit residual budget. The two halves of the paper meet here. In this construction, block ownership did not survive the preceding normalization as usable scaling information; application-unit accuracy does survive it, once it is encoded as a residual budget carried with the row.

5 Residual-budget contracts

The first half of the question asked which generator information explains the numerical gain, and the answer was that block ownership does not. This half asks which generator information can be turned into a guarantee. Ownership is one kind of generator information; a tolerance stated in application units is another. A residual budget says how much violation of an original constraint is acceptable. It must be supplied by the application or declared as a numerical test parameter; it cannot be inferred from a block label or a zero right-hand side. Scaling rows so that solver tolerances correspond to negligible application-unit violations is established practice [11, 15, 17]. The results below make its combination with coefficient limits exact: they characterize every admissible factor, certify when none exists, select the centered one, and identify what row scaling cannot control. Individually the steps are elementary algebra; what the section contributes is that the contract is complete, in the sense that a factor is admissible exactly when the conditions hold, and that each consequence is stated in the units the application uses.

For a nonconstant row $a^T x \leq b$, define the original residual $v(x) = (a^T x - b)_+$ and a budget $\delta > 0$. Let $\epsilon > 0$ bound the residual of the *exported* row. This is a hypothesis about the returned point rather than a consequence of a solver parameter: presolve, internal scaling and finite precision intervene between the two. For a positive row factor α ,

$$(\alpha a^T x - \alpha b)_+ = \alpha v(x) \quad (1)$$

in exact arithmetic, so an exported residual at most ϵ implies $v(x) \leq \delta$ whenever $\alpha \geq \epsilon/\delta$. Absolute residuals give the same identity for equalities. Write $m = \min_{a_j \neq 0} |a_j|$ and $M = \max_j |a_j|$, and let $0 < \ell \leq U$ limit the exported coefficient magnitudes. An additional RHS limit U_b intersects the upper endpoint below with $U_b/|b|$ for $b \neq 0$.

Proposition 5.1 (Admissible factors and constrained centering) *The factors that place every nonzero coefficient in $[\ell, U]$ and satisfy $\epsilon/\alpha \leq \delta$ form exactly the interval*

$$\mathcal{I} = [L, H], \quad L = \max\{\epsilon/\delta, \ell/m\}, \quad H = U/M. \quad (2)$$

The interval is nonempty if and only if

$$M/m \leq U/\ell \quad \text{and} \quad \delta \geq \epsilon M/U. \quad (3)$$

When it is nonempty, the unique minimizer of

$$\min_{\alpha \in \mathcal{I}} \max_{a_j \neq 0} |\log(\alpha |a_j|)| \quad (4)$$

is

$$\alpha^* = \min\{H, \max\{L, (mM)^{-1/2}\}\}. \quad (5)$$

Proof The coefficient bounds are equivalent to $\alpha m \geq \ell$ and $\alpha M \leq U$. Together with $\alpha \delta \geq \epsilon$ they give (2). The condition $L \leq H$ is equivalent to the two inequalities in (3). To solve the objective, put $t = \log \alpha$, $q = (\log m + \log M)/2$ and $h = (\log M - \log m)/2$. Since the extreme absolute logarithms occur at an endpoint,

$$\max_j |t + \log |a_j|| = \max\{|t + \log m|, |t + \log M|\} = h + |t + q|.$$

The unique minimizer on $[\log L, \log H]$ is the projection of $-q$ onto that interval. Exponentiating gives (5).

Incompatibility certificate. When $L > H$, (3) names the violated requirement. A within-row spread $M/m > U/\ell$ cannot be repaired by any row factor; a budget $\delta < \epsilon M/U$ conflicts with the coefficient ceiling even for a perfectly balanced row. In both cases the certificate concerns the numerical specification, not the optimization model: exporting the unscaled row does not meet the contract, and resolving the conflict requires revising units, limits or the formulation. The factor α^* optimizes coefficient centering subject to the contract; it is not a condition-number minimizer.

Corollary 5.2 (Invariance under consistent re-emission) *Replace (a, b, δ) by $(qa, qb, q\delta)$ for any $q > 0$, keeping ϵ, ℓ, U fixed. The interval becomes $\mathcal{I}/q$, abstention is unchanged, and the selected factor becomes α^*/q . The exported row and the budget-normalized original residual $v(x)/\delta$ are therefore unchanged.*

Proof Both coefficient extremes and the original budget acquire the factor q . Each term defining L , H and $(mM)^{-1/2}$ is divided by q . Equation (1) gives the residual statement.

A budget is therefore information that survives preprocessing: generators that emit the same constraint in different units, each with the budget expressed in its own units, obtain the same exported row and the same contract. The invariance is lost if a row is re-emitted without transforming its budget, and a fixed cutoff on the factor itself does not have it. Restricting factors to powers of two [3] retains it for power-of-two re-emissions, subject to the representability conditions of Proposition 5.3.

5.1 The integer-rounding term

Let J be the integer columns and let $\bar{x}$ be obtained by rounding only these coordinates of x . Suppose $|\bar{x}_j - x_j| \leq \eta_j$ for $j \in J$ and put $E = \sum_{j \in J} |a_j| \eta_j$. The triangle inequality and (1) give

$$v(\bar{x}) \leq \epsilon/\alpha + E, \quad (6)$$

also for equalities with absolute residuals. Changing the row factor reduces the exported-residual term but leaves the rounding term, which is measured in original units, unchanged. Normalizing a big- M row therefore cannot remove the effect of an integrality tolerance on that row.

For a contract on the rounded assignment, Proposition 5.1 applies with δ replaced by $\delta - E$ when $\delta > E$; if $\delta \leq E$, no positive factor certifies it. The bound is a worst case: actual rounding errors may be smaller or cancel, and other rows can exclude the worst case, so an empty interval shows that the stated information is insufficient rather than that returned points will violate the budget. Under re-emission E becomes qE , so invariance is retained. A tighter integrality tolerance or a reformulation can reduce E ; row scaling cannot.

5.2 Exact export on a dyadic grid

A factor in the real admissible interval does not guarantee an exactly proportional exported row: the products $\text{fl}(\alpha a_j)$ and $\text{fl}(\alpha b)$ are rounded independently, and rounding a computed endpoint to nearest can leave the closed interval. Our arbitrary-factor implementation forms the endpoints as exact rational numbers in the supplied binary64 inputs, rounds them *inward*, and abstains if no positive binary64 factor remains. This certifies admissibility of the factor, not exactness of the products.

Powers of two avoid multiplication rounding. The following intersection also accounts for subnormal numbers and for the RHS. Assume finite binary64 input data and IEEE arithmetic with gradual underflow [18]. Write each nonzero datum $z \in \{a_1, \dots, a_n, b\}$ uniquely as $z = \text{sign}(z)N_z 2^{p_z}$ with N_z a positive odd integer, and put $q_z = \lfloor \log_2 |z| \rfloor$. Let $[L, H]$ be the desired real factor interval, including any rounding allowance from (6).

Proposition 5.3 (Exact dyadic admissibility) *A factor $\alpha = 2^k$ is positive and representable in binary64, belongs to $[L, H]$, and multiplies every nonzero coefficient and the RHS exactly into a finite binary64 value if and only if the integer k satisfies*

$$K_- = \max \left\{ -1074, \lceil \log_2 L \rceil, \max_z (-1074 - p_z) \right\} \leq k, \\ k \leq K_+ = \min \left\{ 1023, \lfloor \log_2 H \rfloor, \min_z (1023 - q_z) \right\}.$$

If $K_- > K_+$ there is no such factor. Otherwise an exponent nearest to $-\frac{1}{2} \log_2(mM)$ among the integers in $[K_-, K_+]$ minimizes the logarithmic coefficient objective (4) on this exact-export set; a tie may be resolved by choosing the larger factor.

Proof The factor itself requires $-1074 \leq k \leq 1023$. Since the datum's odd significand is unchanged, the product $N_z 2^{p_z+k}$ lies on the subnormal grid exactly when $p_z + k \geq -1074$; its highest bit stays below overflow exactly when $q_z + k \leq 1023$. The original significand has at most 53 bits, so these two conditions are also sufficient. Intersecting them over the row data and with $L \leq 2^k \leq H$ gives the interval of integer exponents. The objective is $\frac{1}{2} \log(M/m) + |k \log 2 + \frac{1}{2} \log(mM)|$, proving the nearest-exponent claim.

The implementation obtains the endpoint exponents from integer ratios and bit lengths, compares candidate objectives exactly as rational numbers, and checks the rounded products against exact rational products. Writing the row to a text file additionally requires a round-trip decimal serializer, since fixed decimal places can discard small values even when the multiplication is exact.

Table 2 Endpoints and how they are measured.

Endpoint	Measurement	Results
Exact equivalence	Whole-row proportionality of named exports, including RHS	§7.1, §7.3
Numerical representation	κ_1 of one fixed basis; κ_2^+ of exported matrices	§7.1, §7.3
Mechanism attribution	Matched exports: byte identity and paired spectral ratios	§7.3
Solver effort	Gurobi work units on runs completed by every policy	§7.4
Failure-sensitive performance	PAR10 over a fixed instance pool	§7.4
Solver acceptance	Solver status and returned point	§7.5
Verified correctness	Exact original residuals, bounds and integrality; enumerated optimum	§7.5

Verification after solving. The contract is conditional on the exported residual, so returned points are checked against the original rows, bounds and integrality restrictions. On the stress instances of Section 6 the check is exact: row residuals are rational expressions in the stored binary64 data, and optima are obtained by enumeration. On the dispatch models the archived checker uses floating-point residuals, which are diagnostics; certifying branch-and-bound results is a stronger task [4].

6 Computational design

The experiments measure the endpoints in Table 2. Each result is stated for its own endpoint and none is used as a proxy for another: a condition number describes a representation, not solver effort or the correctness of a returned point, and a solver status is not an independently verified solution.

6.1 The instance bank

The dispatch models come from one generator, written for the hydrothermal operation of the Uruguayan power system, and they are what that generator produces rather than a selection from a benchmark library. All have a 360-hour horizon and fall into three classes. *Simple* (204 models) dispatches the Río Negro cascade — the reservoirs Bonete, Baygorria and Palmar, in that order downstream — against a fixed demand with a cost of unserved energy. *SG-Ter-Mer* (68 models) dispatches the Uruguayan share of the binational Salto Grande plant together with a thermal unit against a market, whose unserved demand is priced by a piecewise-linear curve; its segments are the *market-shortage rows* referred to below. *Full* (34 models) combines both and values Bonete’s stored water by Bellman cuts. The integer variables are the indicators that select the height band of each reservoir’s production function.

Table 3 The dispatch instance bank, from the unscaled export of every model. Entries are the median and, where it differs, the range over the class. Sizes are constant within a class; decades is $\log_{10}$ of the ratio between the largest and smallest magnitude in a model.

Class	n	Rows	Columns	Nonzeros	Integer vars
Simple	204	14,400	10,800	35,590	2,160
SG-Ter-Mer	68	10,080	6,120	22,679	1,440
Full	34	23,763	16,561	57,555	3,600
Class	n	Coefficient magnitudes		Coef. decades	RHS decades
Simple	204	1.3×10^{-3} to 1.8×10^9		11.9 [11.6, 12.1]	9.0 [8.7, 9.2]
SG-Ter-Mer	68	2.8×10^{-4} to 1.4×10^{10}		13.3 [12.9, 13.6]	8.4 [7.8, 9.0]
Full	34	1.7×10^{-4} to 1.4×10^{10}		13.4 [12.9, 13.8]	9.3 [8.9, 9.4]

Table 3 gives the sizes. Within a class they do not vary: the horizon and the plant set are fixed and the instances differ in their data — inflows, demands, water values, initial volumes — not in their structure. That is a limitation of the bank, and it is why the conditioning audit below reports a per-instance ratio rather than treating the 306 models as a size sweep. What does vary, and is the reason scaling is in question at all, is the numerical range: a single model mixes reservoir volumes in cubic metres, with coefficients reaching 10^9 in Simple and 10^{10} in the other two classes, with power in megawatts and with water values near 10^{-4} , so its coefficient magnitudes span 11.9 decades in the median Simple model and 13.4 in the median Full one, before any scaling. One Simple model, caso46e, is used in Section 7.5 to state residual budgets in application units; it is otherwise unremarkable within its class.

Nine of the SG-Ter-Mer instances carry a defect in their total-demand series, described in Online Resource 1: they remain well-posed MILPs, and every comparison here is between representations of one model, but they are not faithful market scenarios.

6.2 Fixed-basis conditioning audit

A condition number read along each policy’s own solve compares different vertices as well as different representations. This audit holds the vertex fixed. For each of the 306 archived dispatch inputs (204 Simple, 68 SG-Ter-Mer and 34 Full models), the C++ generator exports Base, Flat-GM, Flat-L2, Role-Hybrid and SA-Mat without calling an optimizer. HiGHS 1.15.1 solves the Base LP relaxation with the simplex method on one thread, and its optimal basis is recorded by row and column names. The same basis is then assembled from every policy’s own export: structural basic columns come from the scaled matrix and basic slacks enter as unit columns, as a solver holds them. Before assembly, each export is checked to be a whole-row positive scaling of Base, including RHS values, with identical bounds, objective and integrality (relative tolerance 2×10^{-13}).

The diagnostic is $\kappa_1(B) = \|B\|_1 \|B^{-1}\|_1$, with $\|B^{-1}\|_1$ estimated from a sparse LU factorization by the block 1-norm estimator [13]. Because κ_1 is not invariant under row scaling, the audit measures the conditioning of the exported representation before any solver-internal scaling, not an intrinsic property of the vertex. An independent script rebuilds every basis from the LP text and recomputes each estimate through an exact power-of-two row equilibration, and a precision check bounds κ_1 from both sides on a stress sample. The instance set is complete, and all exports, bases and hashes are archived.

6.3 Matrix attribution audit

A C++ generator of block-structured instances builds local blocks and coupling rows and retains row roles and column ownership, so every policy in Table 1 is available in one implementation and exports can be compared before any optimizer is called. Two generator parameters control the disturbance it introduces. The *severity* S is its half-width in decades: a disturbed row is multiplied by 10^ξ with ξ uniform on $[-S, S]$. The *pattern* says which rows are disturbed and whether they share a draw. Under **coupling** each coupling row draws its own ξ , which is the adversarial case, since no single common factor can compress heterogeneity between rows; under **coupling_uniform** all coupling rows share one draw, which is the collective change of units that a common factor is designed for; under **mixed** local rows are disturbed as well. The archived headline cell has four heterogeneous blocks, coefficient exponent range $[-2, 2]$, pattern **coupling** at severity 6, and 30 seeds. A fresh replication crosses ten seeds (20260909–20260918) with exponent ranges $[-R, R]$ for $R \in \{1, 2, 3\}$, severities $S \in \{3, 6\}$ and the three patterns: 180 instances and 1,080 exports, written with a serializer that preserves binary64 values on readback. Identity claims rest on direct comparisons of named exports. The spectral pseudo-condition number $\kappa_2^+(A) = \sigma_{\max}(A)/\sigma_{\min}^+(A)$ uses a numerical rank fixed per instance from the row-L2-normalized Base (relative threshold 10^{-12}) and shared by all its variants.

6.4 Operational reconstruction

The operational campaign runs Base, Flat-GM, Flat-L2 and Role-Hybrid on 15 fixed instances from each of the Simple and SG-Ter-Mer classes with Gurobi at relative MIP gaps of 1% and 0.1% (seed 42, 16 threads, 1,800-second limit): 240 assigned runs, whose native logs and input hashes were retrieved from the cluster. Its mean PAR10 is the only quantity reported here in units of time, and all 240 logs name one processor and operating system; Online Resource 1, Table S3 states the environment of every campaign. The two subsets are the ones that historical campaign ran, listed in Online Resource 1; they predate this analysis, so they were not chosen after these endpoints were seen. No selection rule was recorded with them, and we therefore claim only that, not

pre-registration. Nine of the 15 SG-Ter-Mer instances, including instance077, carry the scenario defect of Section 6.1; Online Resource 1 repeats the endpoints on the six instances without it. The pre-specified primary endpoint is failure-sensitive performance: the arithmetic-mean PAR10 over all 15 instances of a cell, with 18,000 seconds assigned to a non-completion. Solver effort, the secondary endpoint, is Gurobi’s work measure, recovered from the native logs at a common resolution of 0.01 units, on the instances that every policy completed; paired geometric-mean ratios carry exploratory bootstrap intervals (Online Resource 1). The two gap settings reuse the same instances and are not independent replications.

6.5 Enumerated residual-budget stress experiment

The design follows the idea of Neumaier and Shcherbina [23], who build small mixed-integer programs on which a tolerance-based solver returns an answer that is wrong in exact arithmetic. Here the construction is used not to exhibit a failure but to compare representations of one model under a rule that decides acceptance independently of the solver. Three binary families with 12 variables contain a cardinality row, three positive packing rows and three rows with signed integer coefficients; profits are positive integers. With integer matrix W and integer capacities k , the represented constraints are

$$10^e Wx \leq 10^e k - 0.1 \mathbf{1}, \quad x \in \{0, 1\}^{12}, \quad e \in \{0, 3, 6, 9\}. \quad (7)$$

For these exponents the binary feasible set is exactly $Wx \leq k - \mathbf{1}$, and the stored binary64 right-hand sides are checked to separate the adjacent scaled integer activities. The exponent therefore changes the representation and the continuous relaxation without changing the binary optimum. Because the absolute margin 0.1 is fixed, this is not pure re-emission: it deliberately places infeasible integer points at a controlled distance from the continuous boundary. The budget is 0.01 in the units of the represented row, a declared test parameter.

Eight seeds (100–107), reserved after a one-seed pilot, give 96 models. Each is solved with presolve on and off by HiGHS 1.15.1, Gurobi 13.0.2 and CPLEX 22.1.2 on one thread, with zero target gap, a ten-second limit, and primal and integrality tolerances 10^{-6} . Base, GM and L2 scale every row by their kernels without an identity band; GM-tight uses GM with both tolerances tightened to 10^{-9} ; Budget-only uses the direct factor ϵ/δ_r with $\epsilon = 10^{-6}$; Budget-GM applies Proposition 5.1 with coefficient limits $[10^{-9}, 10^9]$; Budget-round additionally reserves $E_r = 10^{-6} \sum_j |a_{rj}|$ and abstains on the whole model if any row’s interval is empty. Dyadic-GM and Budget-dyadic use power-of-two factors under Proposition 5.3, without and with the budget. These two arms were added after an endpoint rounding issue was found in the initial run and are exploratory; the original seven arms were rerun with the corrected implementation and unchanged seeds. The nine policies give 1,728 planned configurations per solver; Budget-round abstains on 144 of them, leaving 1,584 calls per solver and 4,752 in total.

Table 4 Fixed-basis conditioning audit on all 306 dispatch models. One optimal basis of the Base LP relaxation is assembled from every policy’s own export. The first row is the median κ_1 of the Base basis; the others give the median per-instance ratio $\kappa_1(\text{policy})/\kappa_1(\text{Base})$ and, in parentheses, the number of instances with a ratio below one. Ranges are in Online Resource 1, Table S1.

Policy	Simple ($n = 204$)	SG-Ter-Mer ($n = 68$)	Full ($n = 34$)
Base, median κ_1	8.8×10^{18}	2.9×10^{20}	4.3×10^{20}
Flat-GM	2.4×10^{-7} (204)	7.3×10^{-9} (68)	3.5×10^{-7} (34)
Flat-L2	1.57 (32)	1.51 (10)	1.49 (2)
Role-Hybrid	2.4×10^{-7} (204)	7.3×10^{-9} (68)	3.5×10^{-7} (34)
SA-Mat	2.4×10^{-7} (204)	7.3×10^{-9} (68)	3.5×10^{-7} (34)

A configuration is *verified* when the returned point is within 10^{-9} of a binary assignment that is exactly feasible for the stored binary64 data and attains the optimum found by enumerating all 2^{12} assignments, and the bounds hold within 10^{-9} . Residual budgets and integrality errors are recorded separately, and the rule does not use the solver status. The instances are small and deliberately close to integer feasibility boundaries, so counts describe a failure mechanism rather than practical failure rates, and repeated solver settings on one model are not treated as independent samples. A later control on the same 96 stored models applies GCD division with RHS flooring, an established integer-aware reduction that preserves every binary feasible set but changes the LP relaxation; Online Resource 1 reports it together with two GM variants.

7 Results

7.1 Scaling materially changes numerical representation

All 1,530 exports pass the whole-row scaling check, with largest relative deviation 2.5×10^{-16} , and every policy is evaluated on the same basis. The three policies with GM local normalization lower κ_1 on every one of the 306 models, with median ratios of 2.4×10^{-7} in Simple, 7.3×10^{-9} in SG-Ter-Mer and 3.5×10^{-7} in Full (Table 4). Because the vertex is common, these are differences between representations rather than between the vertices that different solves reach. The independent reconstruction reproduces all 1,530 bases exactly. Because the unscaled values exceed the reciprocal of machine precision, their accuracy was also checked directly on a stress sample of 24 bases from eight models: in each class, the model with the largest Base estimate, the model with the weakest Flat-GM ratio and one seeded random model, under Base, Flat-GM and Flat-L2. A lower bound certified with exact rational residuals and an a posteriori upper bound over all columns of B^{-1} enclose every κ_1 within a relative width of 6×10^{-3} , and the resulting upper bounds on the Flat-GM/Base ratio agree with the estimated ratios to 2.1×10^{-5} (Online Resource 1, Table S2). The median separation of six to eight orders of magnitude,

present on every model, is therefore a property of the representations and not of floating-point error in the unscaled bases. An archived computation on a different reference basis, the LP with integers fixed at a Gurobi incumbent, points in the same direction with smaller magnitudes, but its exports were not retained and it is not used as evidence (Online Resource 1).

The kernel, with the safeguards acting on its candidates, determines this separation. Flat-L2 covers the same rows as Flat-GM, yet its median ratio stays between 1.49 and 1.57 and it lowers κ_1 on only 44 of the 306 models. Every row of these models has a coefficient of magnitude at least one, so the Euclidean factor $1/\|a_r\|_2$ never exceeds one: Flat-L2 enlarges no row, whereas Flat-GM enlarges 22–32% of them, and the identity band and coefficient window leave 58–64% of the rows unscaled under Flat-L2 against 22–30% under Flat-GM. The generator information, in contrast, plays no role here. SA-Mat and Role-Hybrid produce byte-identical exports on all 306 models, and they differ from Flat-GM only on residual market-shortage rows of SG-Ter-Mer and Full that neither covers. Because the coupling rows of these models receive identical factors under Flat-GM, Role-Hybrid and SA-Mat, this identity does not test ownership; the ownership comparison requires the synthetic matrices of the next subsection.

Flat-L2 is worth pausing on, because it is the same kernel that the matched synthetic control will show to be the strong one. On generated dispatch bases it is the weak policy, better than Base on 44 of 306 models; on the synthetic coupling matrices it reproduces the entire advantage attributed to block meta-data. A kernel ranking obtained from one matrix diagnostic therefore does not transfer to another family, still less to solver behavior, and the endpoints of Table 2 have to be measured where they are claimed.

7.2 What the solver’s own scaling leaves

Every solver equilibrates internally, so a conditioning difference measured on the exported file need not be one the solver ever sees. To find out what survives, the same 1,530 bases were rescaled by an iterative row-and-column infinity-norm equilibration with factors rounded to powers of two — the scheme the three solvers describe, and the rounding that keeps every scaled coefficient exact in binary64 — and κ_1 was recomputed after each pass. This is a model of a solver’s scaling stage, not any solver’s implementation, and no optimizer is called; what it isolates is how much of a pre-export difference an equilibration of this kind can remove.

It removes most of it, and it does not remove all of it (Table 5). The equilibration is itself worth ten to fourteen orders of magnitude on the unscaled file: the median Base basis falls from 8.9×10^{18} to 1.6×10^8 in Simple and from 2.9×10^{20} to 2.9×10^6 in SG-Ter-Mer. Of the six to eight orders that separated Flat-GM from Base on the file, what is left afterwards is a factor of about six in Simple and SG-Ter-Mer and about two in Full. The surviving part is not an artifact of precision. On 200 random matrices scaled by exact

Table 5 What survives a solver-style scaling. For each class the first row is the median κ_1 of the Base basis; the others give the median per-instance ratio $\kappa_1(\text{policy})/\kappa_1(\text{Base})$ with, in parentheses, the number of models on which the policy lowers κ_1 . Columns are the exported file and the same bases after two and ten passes of power-of-two row-and-column equilibration.

Class	Policy	n	As exported	2 passes	10 passes
Simple	Base (κ_1)	204	8.9×10^{18}	1.6×10^8	1.6×10^8
	Flat-GM	204	2.4×10^{-7} (204)	0.178 (203)	0.162 (203)
	Flat-L2	204	1.58 (32)	0.448 (173)	0.466 (177)
SG-Ter-Mer	Base (κ_1)	68	2.9×10^{20}	7.7×10^6	2.9×10^6
	Flat-GM	68	7.3×10^{-9} (68)	0.216 (62)	0.181 (68)
	Flat-L2	68	1.51 (10)	1.32 (22)	4.5×10^{-2} (67)
Full	Base (κ_1)	34	4.3×10^{20}	3.2×10^8	2.3×10^8
	Flat-GM	34	3.5×10^{-7} (34)	0.515 (24)	0.514 (24)
	Flat-L2	34	1.51 (2)	0.809 (20)	0.505 (26)

powers of two, so that the scaled matrix is bit-identical to the product and nothing is lost in forming it, a converged equilibration never brought the two representations within a factor of two of each other: the median separation is 22 and the largest 755. An infinity-norm equilibration has no unique fixed point, and a row scaling applied before it is not in its kernel, which is why the file a generator writes still constrains what the solver’s own scaling can reach.

The more consequential observation is that the ranking does not survive. On the exported file Flat-L2 is the weak policy, with median ratios near 1.5 and improvements on 32 of 204, 10 of 68 and 2 of 34 models. After equilibration it improves 177, 67 and 26 of them, and in SG-Ter-Mer its median ratio is 4.5×10^{-2} , four times better than Flat-GM’s 0.181. The policy the file-level audit ranks last is, in the class where the effect is largest, the best one once the solver has scaled. The reverse also happens: Flat-GM, which improves all 34 Full models on the file, improves only 24 of them after equilibration. A condition number read on the exported representation therefore states a property of that representation, as Table 2 says, and it does not order scaling kernels by what a solver will work with — not even in sign, let alone in magnitude.

7.3 A kernel effect mistaken for a block effect

In the archived heterogeneous headline cell, the median spectral diagnostic is 5.30×10^8 for Base, 1.36×10^4 for Flat-GM, 1.33×10^3 for SA-Mat and 4.68×10^2 for Flat-L2. The factor of about 10.2 between Flat-GM and SA-Mat was originally read as the value of block metadata. That comparison also changes the coupling kernel. With the kernel matched, no ownership component remains: Role-Hybrid, which uses no block scales, matches SA-Mat in all 30 archived records, and permuting the block-scale assignment leaves the diagnostic unchanged. Proposition 4.2 gives the reason: after GM local normalization, the block-relative formula is Euclidean normalization. Two

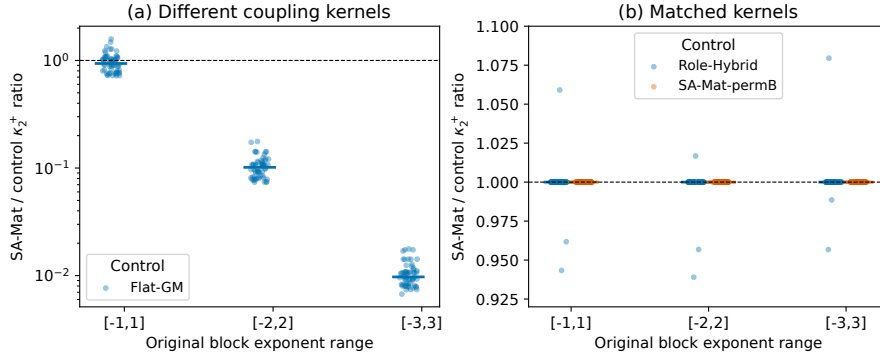


Fig. 1 Fresh matrix audit. Each exponent range contains 60 generated instances. Panel (a) changes the coupling kernel; panel (b) holds the kernels fixed and removes (Role-Hybrid) or permutes (SA-Mat-permB) the block-scale information. Marks are individual ratios with deterministic horizontal jitter; short horizontal segments are medians. The panels use different vertical scales. Unscaled matrices, whose estimates approach binary64 resolution, enter no ratio; all paired scaled diagnostics are below 6.5×10^6 .

further tests agree. Pre-local block scales (SA-Pre) are worse than Role-Hybrid on all 30 instances, with median paired ratio 4.27, and in the exhaustive 24-permutation assignment sweep the true assignment is never the best (median rank 18).

The fresh replication reproduces this attribution on 180 new instances (Figure 1). All 1,080 exports pass the whole-row proportionality check at relative tolerance 10^{-13} . SA-Mat and Role-Hybrid are byte-identical on 136 of the 180 instances, and SA-Mat and its permuted-scale counterpart on 165. Where they differ, the implemented coefficient-window vetoes and filtered block extrema lie outside the hypotheses of Proposition 4.1, and the effect is small: the SA-Mat/Role-Hybrid spectral ratio lies between 0.939 and 1.079 (median one), and the permutation ratio between 0.99962 and 1.00022. By contrast, the median SA-Mat/Flat-GM ratio falls from 0.937 to 0.102 and 0.00973 as the exponent range grows from $[-1, 1]$ to $[-3, 3]$. The large, heterogeneity-dependent advantage is therefore a coupling-kernel effect; ownership information accounts at most for the small, implementation-dependent differences in panel (b). Flat-L2 attains the smallest diagnostic in the archived cell because it also changes local normalization, and on the dispatch bases of Section 7.1 the same kernel leaves conditioning essentially unchanged: kernel rankings on synthetic matrices need not transfer.

7.4 Completed-run gains are not failure-robust

On the 14 SG-Ter-Mer instances that every policy completed, Flat-GM used less solver work than Base, with paired geometric-mean ratios 0.698 at 1% gap and 0.682 at 0.1% (bootstrap intervals $[0.486, 0.944]$ and $[0.470, 0.925]$): about 30–32% less work on that pool. The fifteenth instance, instance077,

completes under Base and times out under every scaled policy at both gaps. Under PAR10 this single failure dominates: Base averages 116 and 134 seconds, while every scaled policy averages between 1,216 and 1,218 seconds (Online Resource 1, Table S5). Scaling therefore produced a sizeable completed-run efficiency gain in this class, but the gain was not robust to failure-sensitive aggregation. On the six instances of the class that are free of the scenario defect the conditional gain is larger and, since instance077 is not among them, the reversal disappears (Online Resource 1, Table S4). The Simple class has no uniform ordering: Flat-GM lowers mean PAR10 from 11.9 to 2.1 seconds at 1% gap and raises it from 52.0 to 73.0 seconds at 0.1%. Role-Hybrid stays within one second of Flat-GM in all four cells.

Replication under a second seed and a second solver. The reconstruction is one solver at one seed. The same 30 instances, four policies, two gaps and 1,800-second limit were rerun changing one factor at a time (Online Resource 1, Table S6). Under Gurobi with seed 7 the SG-Ter-Mer effort ratio of Flat-GM is 0.778 at a 1% gap and 0.777 at 0.1%, against 0.698 and 0.682 at seed 42: the conditional gain survives the change of seed. Under CPLEX at the original seed, in its own deterministic effort metric, the same ratio is 1.078 and 1.077: it does not survive the change of solver. On the Simple class the three conditions agree in direction, Flat-GM below one at the loose gap and between 0.84 and 1.06 at the tight one, Flat-L2 above one in all six cells. The instance that reverses the ordering, instance077, behaves differently in each condition: under Gurobi at seed 7 every policy times out, Base included, while under CPLEX the run that reaches the limit is Base at the 1% gap and Role-Hybrid at 0.1%. The completed-run gain of this class is therefore a property of a representation together with a pipeline, and the failure that reversed the PAR10 ordering is not a stable property of the instance either. Neither observation is surprising in the light of what is known about performance variability in mixed-integer programming: Lodi and Tramontani [21] document changes of this order from perturbations that leave the model mathematically unchanged, such as a different seed or a permuted formulation. A row scaling is exactly such a perturbation. The seed-to-seed movement reported here, from 0.698 to 0.778, is of the same order as the effect being measured, which is why these numbers are reported as a measured conditional gain on a fixed pool rather than as an estimate of acceleration.

The two endpoints answer different questions: completed-run effort describes how a representation affects the runs that finish, and PAR10 whether it changes which runs finish. For the reconstructed campaign itself, with one seed and 15 fixed instances per class, the result is a measured conditional gain, not a population estimate of acceleration; the two sensitivity reruns reinforce that caution rather than providing population-level replication. The same runs also show why a stated budget is needed to read residuals: an absolute residual of 4.97 on a zero-RHS equality of caso46e has an activity-normalized value of 1.15×10^{-7} , and without an application budget neither number establishes acceptance; Table 7 supplies one.

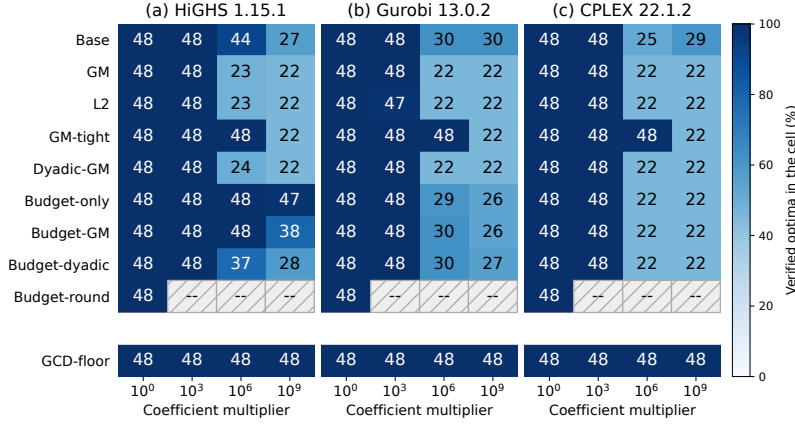


Fig. 2 Verified optima by coefficient multiplier. Each cell counts the configurations whose returned point is a verified optimum, out of the 48 in that cell (three families, eight seeds, two presolve settings); the shading is that count as a percentage, on the common scale at the right. The fixed margin in (7) brings the continuous feasibility boundary increasingly close to an integer activity. Hatched cells marked – are abstentions, where the policy declines to export. The detached bottom row is GCD-floor, the later integer-aware reformulation control, not a scaling policy.

Table 6 Paired verification outcomes on identical model, presolve and solver settings (192 per comparison and solver). “A only” counts configurations verified under A and not under B.

Comparison A vs B	HiGHS		Gurobi		CPLEX	
	A only	B only	A only	B only	A only	B only
Budget-GM vs GM	41	0	12	0	0	0
Budget-dyadic vs Dyadic-GM	19	0	13	0	0	0
Budget-only vs GM	50	0	11	0	0	0
GM vs Base	0	26	0	16	0	10
Budget-only vs Base	24	0	1	6	0	10
Budget-GM vs Base	15	0	1	5	0	10
Budget-dyadic vs Base	2	8	1	4	0	10
GM-tight vs GM	25	0	26	0	26	0
Budget-only vs GM-tight	25	0	4	19	0	26
Budget-only vs Budget-GM	9	0	0	1	0	0

7.5 Budgets improve verification; acceptance is pipeline-dependent

These models are small, adversarial and built so that an integer point sits a fixed distance from the continuous boundary, so the counts below compare policies on a deliberately hard family; they are not estimates of how often a solver fails in practice.

The residual budget has a consistent effect within each scaling rule (Table 6). Plain GM centering loses verified optima relative to Base on all three solvers, 26 configurations on HiGHS, 16 on Gurobi and 10 on CPLEX, with no gains. At multipliers 10^6 and 10^9 it shrinks the fixed 0.1 margin of (7) below the solver

tolerance, and most of its failures (49 of 51 on HiGHS, 36 of 52 on Gurobi, 34 of 52 on CPLEX) return points that satisfy the exported tolerance while violating the original row by more than the budget. Adding the budget to the same rule never lost a verified configuration on any solver. It recovered 41 of them on HiGHS and 12 on Gurobi, and none on CPLEX, where a rule and its budgeted version verify exactly the same configurations; the dyadic pair shows the same pattern, 19, 13 and 0. The two comparisons say different things about the same factors. Plain centering improves the numbers in the exported file and loses verified optima; the same rule with a budget attached loses none. A representation can be numerically better without being operationally safer, unless what safer means is stated in the units of the original model.

Relative to the unscaled model, the outcome depends on the pipeline: presolve, internal scaling, tolerances and finite precision all intervene between the exported row and the accepted point. The complete counts for every policy and solver are in Online Resource 1, Table S7. On HiGHS, Budget-only verifies 191 of 192 configurations and Budget-GM 182, against 167 for Base and 141 for GM; Budget-only gains 24 configurations over Base and loses none. On Gurobi, Budget-only verifies 151, Budget-GM 152 and Base 156. On CPLEX every scaled policy verifies 140 and Base 150: the budget changes nothing there, and the ten configurations lost to scaling are lost with it and without it. On both commercial solvers, tightening the two tolerances to 10^{-9} (GM-tight, 166 configurations on each of the three solvers) is the more effective intervention. The same stored models and row factors thus lead to different acceptance in three pipelines, and the budget helps where this experiment shows the greater sensitivity to the exported coefficients.

The gap between Budget-only and Budget-GM on HiGHS is informative in its own right. At multipliers 10^6 and 10^9 , the only ones at which their outcomes differ, the budget endpoint is binding and Budget-GM selects that endpoint rounded inward, one unit in the last place (ulp) away from Budget-only's factor. All ten discordant configurations, nine on HiGHS and one on Gurobi, with none on CPLEX, have exactly this one-ulp difference. The nine-configuration gap therefore measures how sensitive solver acceptance is to the last bit of a factor near a tolerance boundary, not an effect of coefficient centering. The same sensitivity appears at a larger scale: Budget-dyadic, which uses the nearest admissible power of two and exports every product exactly, verifies 161 configurations on HiGHS, 153 on Gurobi and 140 on CPLEX. Exact representation of the scaled row does not by itself secure acceptance.

Two controls delimit the result. GCD division with RHS flooring verifies all 192 configurations on each solver: it removes the deliberately narrow integer margin, whereas a budget controls the original-unit meaning of an exported residual, so the budget result is a comparison among scalings and not an advantage over integer-aware presolve. Budget-round, which reserves the worst-case rounding term of (6), verifies all 48 configurations it accepts per solver and abstains on the other 144, including many that other policies solve correctly. Its abstentions certify that the requested uniform contract cannot be asserted; they are not solved instances, and Figure 2 counts them separately.

Table 7 A declared residual contract on the dispatch model, applied to the 4,320 rows of caso46e whose units are interpretable. Budgets are stated by the modeller, not measured or supplied by the operator. Admissible factors are the interval of Proposition 5.1 for $\epsilon = 10^{-6}$ and coefficient limits $[10^{-9}, 10^9]$. The geometric-mean factor is constant within every group except the tangents, where it ranges over $[1.99, 2.33]$; the column “GM (med.)” gives its median, and “Worst resid.” gives ϵ divided by the *smallest* factor of the group, the largest original-unit residual the tolerance allows there, in the units of that group’s budget. The two therefore coincide except on the tangents, where the worst residual 5.0×10^{-7} belongs to the row with factor 1.99 rather than to the median row, whose residual is 4.7×10^{-7} . The last group is written pre-scaled, with a coefficient of 10^7 on the generation variable, so 0.1 MW there is 10^6 in the units of the row as written. The admissibility check covers the three kernels the four policies use — the unit factor, the geometric-mean factor shown here and $1/\|a_r\|_2$ — and every one of them is admissible on every row. The archived admissibility check records the corresponding counts for each kernel.

Constraint group	Budget δ	Admissible factors	GM (med.)	Worst resid.
Generation identities, demand cap	0.1 MW	$[10^{-5}, 10^9]$	1	10^{-6}
Generation definition, Bonete/Palmar	0.1 MW	$[10^{-5}, 10^9]$	1	10^{-6}
Production envelope tangents	0.1 MW	$[10^{-5}, 10^9]$	2.14	5.0×10^{-7}
Water balance, Bonete/Palmar	10^3 m^3	$[10^{-9}, 2.8 \times 10^5]$	0.0167	6.0×10^{-5}
Water balance, Baygorria	10^3 m^3	$[10^{-9}, 2.8 \times 10^5]$	2.8×10^{-4}	3.6×10^{-3}
Generation definition, Baygorria	10^6 row units	$[7.0 \times 10^{-8}, 100]$	2.6×10^{-3}	3.8×10^{-4}

The same contract on the application. The 0.01 of the stress family is a test parameter, but the dispatch rows carry interpretable units, so a budget can be stated in them. For illustration, suppose the modeller declares 0.1 MW for the rows that account for generation and cap it by demand, 0.1 MW for the piecewise-linear production envelopes, and $1,000 \text{ m}^3$ — a thousandth of a cubic hectometre — for the reservoir water balances. Table 7 applies Proposition 5.1 to the 4,320 rows of caso46e that carry those meanings, with $\epsilon = 10^{-6}$ and the coefficient window $[10^{-9}, 10^9]$. The admissible interval is wide, the geometric-mean factor lies inside it on every row, and the residual the tolerance allows in original units stays five to nine orders of magnitude below the declared budget. On this model the contract leaves the geometric-mean factor unchanged on every row: it restricts which factors are admissible, as it must, but the geometric-mean choice considered here already lies well inside that restriction. The budgets are declared rather than measured, so the margin is also the sensitivity: scaling all six by a common multiplier, all three kernels stay admissible on all 4,320 rows down to 10^{-4} and the two the comparison turns on down to 10^{-5} , which places the result far from the chosen numbers (Online Resource 1, Table S8).

Its content is in the reading of a residual. One family is written pre-scaled: the Baygorria generation definition carries a coefficient of 10^7 on the generation variable, so the same 0.1 MW becomes a budget of 10^6 in the units of that row. It is the family on which the accepted Flat-L2 run of Section 7.4 leaves its largest residual, 4.97, which against the row’s own budget is a generation error of 5×10^{-7} MW. The residual as written and the quantity it stands for differ by the seven orders of magnitude of that coefficient, and only a budget carried with the row distinguishes them.

8 Discussion

8.1 What we learned

Row scaling changes the exported representation, and part of that reaches the solver. With the basis held fixed, the representation alone moves the condition number of the exported file by six to eight orders of magnitude in the median, and the kernel decides whether it does. That figure describes the file. A solver-style equilibration applied to the same bases recovers most of it — ten to fourteen orders on the unscaled model — and what the choice of export is still worth afterwards is a factor of about six in two classes and two in the third. The pre-export choice is not erased, because an infinity-norm equilibration does not undo a row scaling, but it is worth far less than the file-level number suggests, and on one class it inverts: the kernel that ranks last on the file ranks first after the solver has scaled.

The mechanism is simpler than the metadata explanation. Under exact GM local normalization, the extrema-based post-local block statistic carries no block-scale information (Proposition 4.1), so the block-relative rule becomes Euclidean normalization of the coupling rows. Separately, on a locally whitened triangular structure the optimal common factor is $(1+\rho^2)^{-1/2}$ (Proposition 4.3); the two coincide when the structure has a single coupling row.

Matching kernels is essential for attribution. The original comparison changed the coupling kernel together with the information used, and the fixed-basis audit shows how much a kernel alone can change: identical coverage, representations orders of magnitude apart. A claim that generator information improves preprocessing needs a kernel-matched control.

A condition number of an exported file does not rank scaling policies. It is the natural thing to measure, because it needs no optimizer and no license, and this paper measures it on 306 models. What Section 7.2 shows is that it should not be used to choose between kernels: after the kind of scaling every solver applies, the ordering it gives is wrong in sign on one of the three classes and compressed by six orders of magnitude on the others. Anyone comparing pre-export scalings by a matrix diagnostic should report what remains of the difference once that diagnostic is recomputed through the solver’s own scaling stage, which costs no optimizer time.

Residual budgets are a meaningful invariant quantity. A residual budget carried consistently with its row defines a contract that is invariant under consistent re-emission — the number changes with the row, the meaning does not — and determines the admissible factors exactly. In the stress test, measured against the corresponding unbudgeted rule, it repaired the verification losses of coefficient centering on two of the three solvers and cost nothing on the third; measured against the unscaled model the outcome still depends on the pipeline.

On the dispatch model the same contract is easy to satisfy — all three kernels the four policies use are admissible on each of its 4,320 interpretable rows, with orders of magnitude to spare — and its work is in the reading of a residual, which without a stated unit says nothing. GCD division with RHS flooring verified every stress configuration, but it answers a different question: it is an integer-aware reformulation that exploits the activity lattice of integer rows, whereas a residual budget is a contract in original units that also applies to continuous rows and to constraints admitting no such reduction.

Solver acceptance and effort are properties of the pipeline. The same representation reduced completed-run work and caused a timeout, the same factors were accepted differently by the three pipelines, and a one-ulp change in a factor changed outcomes in one of them. Pipeline, not solver: presolve, internal scaling, tolerances and finite precision all sit between the exported row and the point that comes back, and the experiments here cannot separate their contributions.

8.2 Scope and limitations

The operational evidence comes from one application: two instance classes and 15 fixed instances per class, from the reconstructed campaign of Section 6.4. The two sensitivity reruns, under a second seed and a second solver, are enough to show that at least one finding is configuration-dependent rather than a stable property of the representation, and far too small to describe the population of configurations. The stress instances are small and deliberately adversarial; their budget of 0.01 is a numerical test parameter, and the dispatch budgets of Table 7 are stated by us to show what the contract implies, not supplied or validated by the operator. Solver outcomes depend on the solver, its version and its settings. The collapse result concerns the extrema-based block statistic after GM normalization; other block statistics, other kernels and physical scale information are outside it, and on the operational models the coupling rows receive the same factors under every GM-local policy, so ownership is tested only on synthetic matrices. Finally, row scaling cannot repair within-row heterogeneity beyond the limit in (3) or the integer-rounding amplification in (6).

9 Conclusion

External row scaling changes the exported representation a great deal and the representation the solver works with rather less. Holding one basis fixed, it moves the condition number of the file by six to eight orders of magnitude in the median while preserving the MILP in exact arithmetic, and the kernel determines whether this happens; after the kind of equilibration a solver applies, what the choice is still worth is a factor of a few. The difference is not erased —

a row scaling is not in the kernel of an infinity-norm equilibration — but the file-level figure is not a measure of what reaches the solver, and it does not even order the kernels correctly.

Correct attribution changes the explanation of the gain. The advantage originally attributed to block metadata is a coupling-kernel effect: exact GM local normalization erases the extrema-based post-local block statistic, and the closed-form coupling factor of the idealized triangular model, which reduces to Euclidean normalization in the single-row case, provides a spectral explanation for it that uses no ownership information.

Residual budgets provide a constructive way to encode information that keeps its meaning in original application units. They come with exact feasibility and incompatibility conditions, a centered factor, invariance under re-emission, and explicit rounding and binary64 export conditions, and in the stress experiment adding the residual budget to the same scaling rule never reduced verification against that same unbudgeted rule, on any of the three solvers tested.

What a solver does with a scaled representation still has to be measured, and its output still has to be verified. The practical lesson is therefore not to avoid external scaling, but to make it accountable, in three steps that cost no optimizer time: attribute its gains with kernel-matched controls, measure them again through the solver’s own scaling stage before believing the ordering they give, and state its accuracy contract in the model’s own units so that solver output can be verified against it.

Data and software availability

The accompanying repository contains the C++ generator, frozen historical aggregates, the reconstructed operational logs and analysis, the fixed-basis audit with its exports, bases and independent verification, the exact-enumeration experiment, the row-factor implementation, tests, and the scripts that generate the tables and vector figures. The current research sources are distinguished from the historical draft, whose positive block-attribution claim is superseded. All newly reported data are stored under **results-revision/research-audit/**. No commercial solver is required to regenerate the figures, rerun the fixed-basis audit, verify the saved binary solutions, or run the HiGHS stress arm. Rerunning the Gurobi and CPLEX arms requires the corresponding licenses.

The repository is at <https://github.com/MathiasRodriguezCastro/structure-aware-row-scaling>. The state reviewed here is the fixed release **v1.0-mpc-submission**, which carries three archives: the manuscript sources, the source and data archive that is Online Resource 2, and the bulk inputs of the fixed-basis audit, 5.18 GB of exported LP text and bases in a 337 MB compressed file. Only the rebuild of all 1,530 bases from that text needs the last one; every number reported here follows from the summaries in Online Resource 2.

Supplementary information

Online Resource 1 provides the triangular-model proof, the fixed-basis audit protocol and complete ranges, operational reconstruction details, frozen experiment protocols, arithmetic implementation checks, and reproduction instructions. The accompanying source and data archive is Online Resource 2.

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